Renfei Wu

linkedin.com/in/renfei-wu
github.com/chr1sren
+1 437-484-0131
chrisren.wu@mail.utoronto.ca

Education

University of Toronto — Toronto, ON, Canada
Bachelor of Science in Computer Science and Applied Mathematics — *Sep. 2023 – Expected Apr. 2027*

- **Cumulative GPA**: 3.93/4.00, **Dean's List Scholar** (2025, 2026)
- **Relevant Coursework**: Deep Learning; Reinforcement Learning; Fundamentals of Robotics; Continuum Robotics; Computer Vision; Algorithm Design; Software Engineering; PDEs; Dynamical Systems; Differential Forms; Statistics

Research Experience

Undergraduate Researcher | *Continuum Robotics, RL/IL, Sim-to-Real* — Sep. 2026 – Present
Continuum Robotics Lab, University of Toronto Mississauga (Supervisor: Prof. Jessica Burgner-Kahrs) — *Mississauga, ON*

- Developing a sim-to-real pipeline for visuomotor policies based on reinforcement and imitation learning on Helix, a fixed-base, three-segment continuum robot.
- Conducting ablation studies on auxiliary-loss regularization of the policy's latent space and its effect on generalization.

Undergraduate Researcher | *Reinforcement Learning, State Representation Learning* — Apr. 2026 – Aug. 2026
MedCVR Lab, University of Toronto Mississauga (Supervisor: Dr. Amey Pore) — *Mississauga, ON*

- Proposed an action-aware bisimulation metric incorporating inverse dynamics and established its theoretical properties, including contraction, uniqueness of the fixed point, and a value-function bound.
- Outperformed prior bisimulation methods on DeepMind Control and ManiSkill manipulation tasks, including visual-distractor and generalization settings; analyzed the method's limitations against augmentation-based baselines in a first-author report.
- Presented a poster at the Toronto Robotics Conference and received the **3rd Prize** poster award.

Undergraduate Researcher — Sep. 2025 – Mar. 2026
Research Opportunity Program, University of Toronto Mississauga (Supervisor: Prof. Tingting Zhu) — *Mississauga, ON*

- Conducted research on the adoption of generative AI in computer science education.
- Developed a prompt-engineered LLM-based AI bot, designed its evaluation experiment, and authored the final report.

Research Assistant — May 2025 – July 2025
China Academy of Information and Communications Technology — *Beijing, China*

- Researched the mathematical foundations of post-quantum cryptography, including Shor's algorithm and the lattice- and code-based NIST PQC finalists, and produced technical reports for the team.
- Co-developed a proof-of-concept IP backbone network mapping system using Flask, SQLite, and Leaflet.js.

Teaching Experience

Teaching Assistant — Jan. 2026 – Present
Department of Mathematical and Computational Sciences, University of Toronto Mississauga — *Mississauga, ON*

- **MAT223 Linear Algebra I** (Fall 2026): Lead weekly tutorials and guide students through problem solving.
- **CSC373 Algorithm Design and Analysis** (Fall 2026): Design marking schemes, grade exams, and invigilate.
- **CSC263 Data Structures and Analysis** (Winter 2026): Graded assignments and exams.

Projects

Undergraduate Data Challenge | *Python, scikit-learn, NumPy, Pandas* — Mar. 2025

- Preprocessed textual survey data, engineered features, and performed model selection and error analysis.
- Achieved 90% test accuracy, ranking 3rd out of 80+ undergraduate teams.

Online Tech Support System | *Python, LLMs, RAG* — Oct. 2024

- Developed an online tech support system in collaboration with the local social organization "Let's Get Together", integrating LLMs with Retrieval-Augmented Generation to answer FAQs automatically.
- Reduced response time from 3–4 days to same-day; pilot tested with 30+ participants, achieving 85% positive feedback.

Pseudocode Compiler | *C++, Lex/Yacc* — Dec. 2023

- Built a C++ compiler (Lex/Yacc) to execute and debug pseudocode from the Cambridge A-Level high school CS curriculum.

Technical Skills

Programming: Python, C/C++, Java, SQL, Bash, JavaScript, HTML/CSS
Frameworks & Tools: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, OpenCV, MuJoCo, ManiSkill, ROS, Git, Docker